

Chinese Remainder Theorem (CRT)

Module I

CRT

- Let $n_1, n_2, \dots, n_k$ be pairwise coprime (that is $\gcd(n_i, n_j) = 1, i \neq j$).
- Then the system of k equations,
 - $x \equiv a_1 \pmod{n_1}$
 - $x \equiv a_2 \pmod{n_2}$
 -
- $x \equiv a_k \pmod{n_k}$ has a unique solution for x module N.

- **Example1:**

- $x \equiv 2 \pmod{3}$
- $x \equiv 3 \pmod{4}$
- $x \equiv 1 \pmod{5}$ we can find x. here every pairwise numbers are co-primes.

Formula for finding x in CRT

$$x = a_1 N_1 c_1 + a_2 N_2 c_2 + \cdots + a_k N_k c_k \pmod{N}$$

• Example:

- $x \equiv a_1 \pmod{n_1}$
- $x \equiv a_2 \pmod{n_2}$
- $x \equiv a_3 \pmod{n_3}$
- formula to compute $N = n_1 * n_2 \dots n_k$
- $N_i = \frac{N}{n_i} \Rightarrow N_1 = \frac{N}{n_1} = \frac{n_1 * n_2 * n_3}{n_1} = n_2 * n_3$
- $N_2 = \frac{N}{n_2} = \frac{n_1 * n_2 * n_3}{n_2} = n_1 * n_3$
- $N_3 = \frac{N}{n_3} = \frac{n_1 * n_2 * n_3}{n_3} = n_1 * n_2$

• Example:

- $x \equiv 2 \pmod{3}$
- $x \equiv 3 \pmod{4}$
- $x \equiv 1 \pmod{5}$
- $n_1=3, n_2=4, n_3=5. N=60.$
- $a_1=2, a_2=3, a_3=1.$
- $N_1 = 20, N_2 = 15, N_3 = 12.$
- $N_1 c_1 = 1 \pmod{n_1} \rightarrow N_1 c_1 \pmod{n_1} = 1$
 $\rightarrow 20 c_1 \pmod{3} = 1$
 $\rightarrow 20/3 = 2, 2 c_1 \pmod{3} = 1$ (by trail $c_1=2$)
 $\rightarrow 2 * 2 \pmod{3} = 1 \rightarrow 4 \pmod{3} = 1.$

$c_1, c_2, c_3 \dots$ are multiplicative Inverses

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• Example:

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- $x \equiv 3 \pmod{4}$
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- $n_1=3, n_2=4, n_3=5. N=60.$
- $a_1=2, a_2=3, a_3=1.$
- $N_1 = 20, N_2 = 15, N_3 = 12.$
- $N_1 c_1 = 1 \pmod{n_1} \rightarrow N_1 c_1 \pmod{n_1} = 1$
 $\rightarrow 20 c_1 \pmod{3} = 1$
 $\rightarrow 20/3 = 2, 2 c_1 \pmod{3} = 1$ (by trail **c1=2**)
 $\rightarrow 2 * 2 \pmod{3} = 1 \rightarrow 4 \pmod{3} = 1.$

$$\begin{aligned} N_2 c_2 &= 1 \pmod{n_2} \rightarrow N_2 c_2 \pmod{n_2} = 1 \\ &\rightarrow 15 c_2 \pmod{4} = 1 \\ &\rightarrow 15/4 = 3, 3 c_2 \pmod{4} = 1 \\ &\text{(by trail **c2=3** that gives remainder 1)} \\ &\rightarrow 3 * 3 \pmod{4} = 1 \rightarrow 9 \pmod{4} = 1. \end{aligned}$$

$$\begin{aligned} N_3 c_3 &= 1 \pmod{n_3} \rightarrow N_3 c_3 \pmod{n_3} = 1 \\ &\rightarrow 12 c_3 \pmod{5} = 1 \\ &\rightarrow 12/5 = 2, 2 c_3 \pmod{5} = 1 \\ &\text{(by trail **c3=3** that gives remainder 1)} \\ &\rightarrow 2 * 3 \pmod{5} = 1 \rightarrow 6 \pmod{5} = 1. \end{aligned}$$

Now compute

$$x = 2 \cdot 20 \cdot 2 + 3 \cdot 15 \cdot 3 + 1 \cdot 12 \cdot 3 \pmod{60} = 251 \pmod{60} = \mathbf{11}$$

CRT

- Example2:

- $x \equiv 3 \pmod{5}$
- $x \equiv 1 \pmod{7}$
- $x \equiv 6 \pmod{8}$

- Compute Answers

- $n_1=5, n_2=7, n_3=8. N=280.$
- $a_1=3, a_2=1, a_3=6.$
- $N_1 = 56, N_2 = 40, N_3 = 35.$
- $N=280.$
- $c1=1, c2=3, c3=3.$
- $X=78$